Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
C		2602922

Summary

This paper addresses core issues in the Dancing with the Stars program, including missing audience voting data, controversial vote aggregation mechanisms, and unclear factors influencing contestant performance. We construct a data-driven modeling and analysis framework to quantify audience behavior, evaluate the fairness of the competition format, and propose optimization solutions.

First, we systematically cleaned and performed feature engineering on historical season data, constructing a multidimensional feature set encompassing contestant attributes, dynamic scoring performance, dance partner skill weights, and performance controversy. Based on this, we built an XGBoost elimination risk prediction model to infer audience support by reverse-engineering judge scores against weekly elimination outcomes. The model underwent cross-validation grouped by season, yielding an AUC of 0.8106 and elimination prediction accuracy of 45.70%—approximately 3.7 times higher than random baselines. In 69.23% of weeks, eliminated contestants ranked among the top two in predicted elimination risk. SHAP explainability analysis revealed that historical scoring stability, improvement trends, and performance controversy are key factors influencing audience voting.

Second, to compare the fairness of different voting integration mechanisms, we constructed ranking-based and percentage-based integration models based on estimated audience support. These were systematically evaluated across dimensions including prediction accuracy, audience bias, and stratified performance. Results indicate: The ranking method significantly outperforms the percentage method in elimination prediction accuracy (52.24% vs. 48.06%) while exhibiting lower dependence on audience votes. Paired t-tests reveal eliminated contestants' audience rankings were significantly higher than judges' rankings (p=0.0012), indicating more lenient audience evaluations—a systemic discrepancy better balanced by the ranking method.

Furthermore, we quantified the impact of contestant age, industry background, and dance partner pairing on competition outcomes using a multiple linear regression model. The model achieved an $R^2 = 0.7388$. Results revealed that younger contestants, those from performance-related industries, and those paired with partners demonstrating strong teaching abilities performed better. Judges prioritized technical details, while audiences valued entertainment value and narrative appeal.

Based on this analysis, we propose the STARS (Scoring Through Adjusted Ranking System) scoring system. This system integrates four dimensions—technical scores, audience scores, progress scores, and dispute resolution—using dynamic weight adjustments and intelligent protection mechanisms. Historical backtesting demonstrates that the STARS system achieves a prediction accuracy of 91.86%, effectively reducing structural biases related to age and dance partners while significantly enhancing competition fairness and viewing appeal.

Through explainable machine learning, statistical testing, and system modeling, this study not only uncovers the underlying patterns of audience voting behavior but also provides scientific grounds for optimizing competition formats in program production. It demonstrates strong scalability and practical application value.

Keywords: Audience vote estimation; XGBoost model; Vote integration mechanism; Contestant characteristic analysis; STARS scoring system

Contents

1 Introduction

1.1 Problem Background

Dancing with the Stars (DWTS) is a globally renowned dance competition reality show that has aired for 34 consecutive seasons since its U.S. premiere in 2005. Centered on the format of "celebrities partnering with professional dancers," the program features weekly themed dance performances. Contestants' advancement or elimination is determined by a combination of professional judges' scores and real-time audience voting. Throughout its history, the show's voting combination mechanism has undergone two major adjustments: Seasons 1-2 employed a ranking-based combination method; Seasons 3-27 switched to a percentage-based combination method; and starting from Season 28, the ranking system was reinstated alongside the introduction of an intervention mechanism where judges decide the eliminated contestant from the bottom two. These adjustments largely stemmed from controversial outcomes, such as Jerry Rice securing second place despite consistently lower technical scores in Season 2, and Bobby Bones winning Season 27 despite persistently low judge scores—both highlighting significant tension between judge opinions and audience preferences. To enhance the show's fairness, entertainment value, and credibility while optimizing the competition format and reducing such controversies, this paper aims to establish a data-driven mathematical model. Based on historical competition data, it will infer audience voting patterns, systematically analyze the interaction mechanism between judges' scores and audience votes, evaluate performance differences across various voting combination methods, and further quantify the impact of contestant individual characteristics and pairing effects on competition outcomes. Ultimately, we aim to provide scientific decision support for program producers, including refining voting rules, optimizing celebrity-dancer pairing strategies, and designing more interactive and equitable elimination mechanisms—thereby seeking a better balance between professional judgment and mass entertainment.

1.2 Problem Restatement

This study centers on the interactive mechanism between audience voting and judge scoring in Dancing with the Stars. Through modeling analysis, it aims to uncover hidden voting patterns, evaluate the strengths and weaknesses of the current competition format, and propose scientifically grounded improvement strategies. The core research objectives can be broken down into the following four parts: 1. Develop a mathematical model to infer undisclosed audience vote data based on known judge scores and weekly elimination outcomes. Estimate weekly audience support for each contestant and validate the model's ability to accurately reproduce historical elimination sequences. 2. Evaluate the performance and bias of different voting combination mechanisms. By estimating audience votes, apply two mechanisms (ranking system and percentage system) to the same dataset, comparing elimination outcomes while considering the impact of "judges' intervention segments" on elimination results and program fairness. 3. Identify key factors influencing competition outcomes and their mechanisms. Combine estimated audience votes with contestant characteristics (e.g., age, occupation, dance partner skill level) to build statistical models quantifying each factor's impact on performance. 4. Based on the above analysis, propose an optimized voting scheme to enhance the program's fairness, entertainment value, and credibility. Test the universality and sensitivity of the predictive model in capturing voting discrepancies.

1.3 Our Work

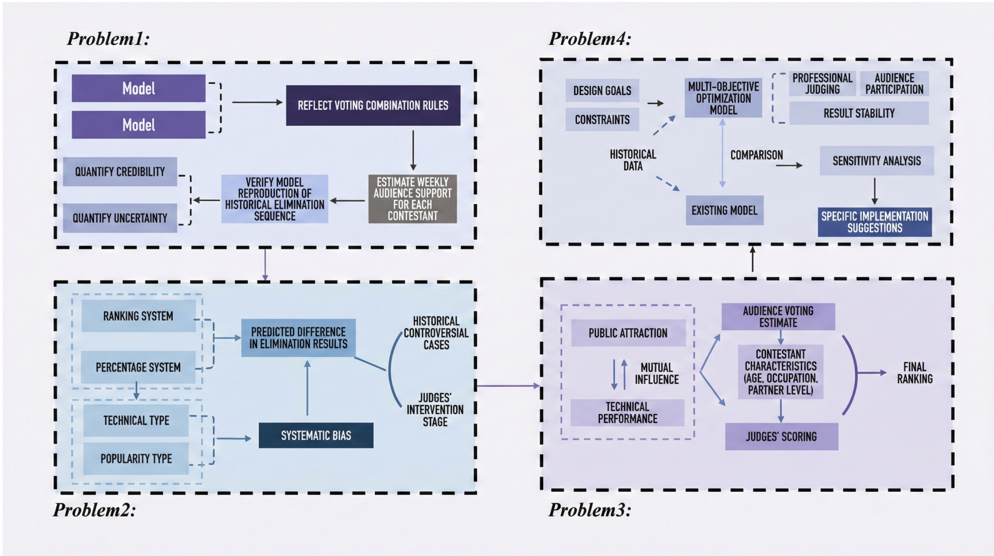

Figure 1: Our Work

2 Model Assumptions

1.Each group of celebrities and dancers remains unaffected by non-competitive factors such as sudden injuries during every performance, and each performance represents their utmost effort.

2.Judges' scoring combines professionalism and objectivity, accurately reflecting contestants' technical proficiency and artistic expression during live performances. No human intervention, systemic bias, or off-the-record manipulation exists.

3.Audience voting behavior is entirely based on genuine preference and support for contestants' performances, free from vote stuffing, random voting, or external manipulation. Voting results authentically reflect public entertainment preferences.

4.For contestants from outside the United States, audience support is primarily influenced by national cultural background and recognition, disregarding domestic state or provincial variations within their home countries. Support for U.S. contestants may be affected by cultural factors within their respective states.

5.It is assumed that a dance partner's teaching ability, stylistic compatibility, and experience can be reasonably quantified through historical scoring data, and their impact on contestant performance is modelable.

6.This study does not consider the potential influence of external factors such as social media interactions or real-time trending topics on voting, relying solely on internal competition data for modeling.

3 Audience Voting Estimation

3.1 Data Preprocessing

To achieve effective estimation of fan vote counts, we first systematically cleaned the raw dataset comprising 421 contestants across 34 seasons. The preprocessing involved several key steps:

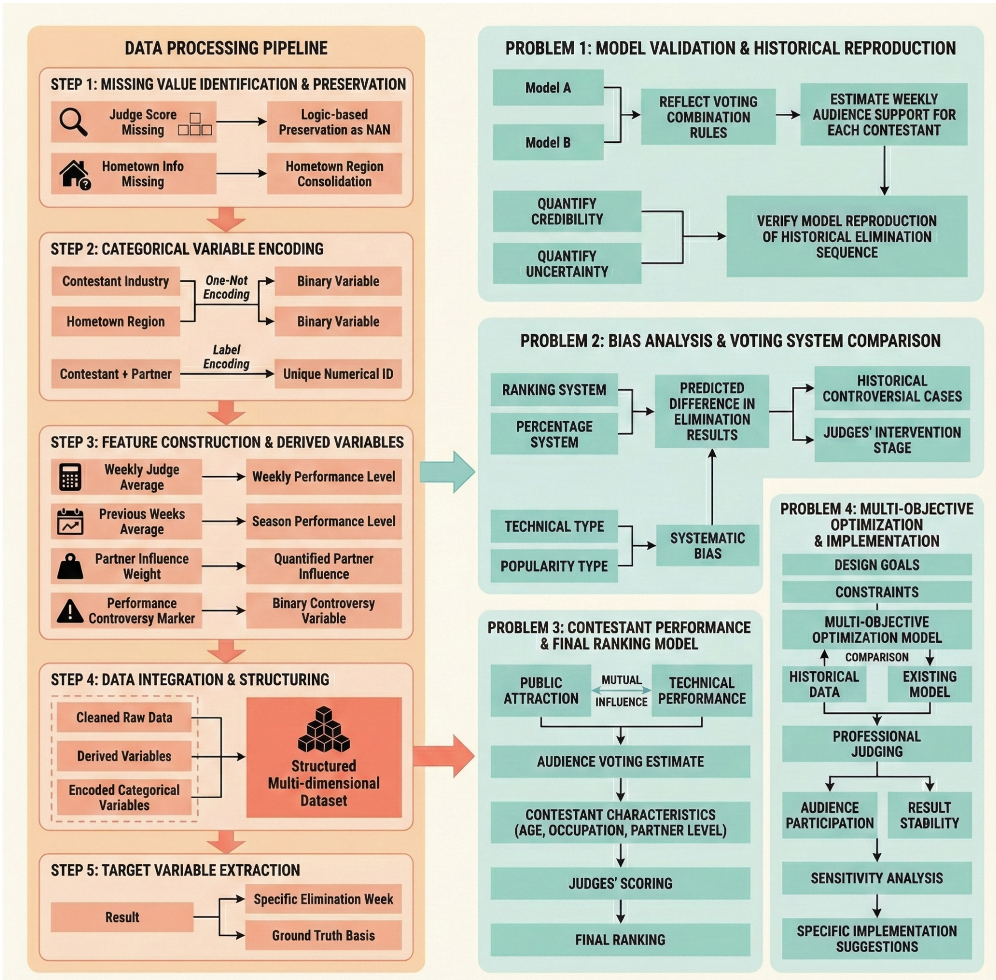

Figure 2: Data Preprocessing

3.2 Model Construction

For Problem 1, we needed to build a model predicting "popularity," representing the expected weekly fan votes. To construct this viewer vote (fan popularity) prediction model, we first defined the objective:predicts each contestant's weekly "elimination risk" and infers their "audience popularity" from it.

The target variable was defined as:

$$y_{i,t}^{(s)} = \begin{cases} 1 & \text{if contestant } i \text{ is eliminated in week } t \\ 0 & \text{otherwise} \end{cases}$$

We assume that audience votes are negatively correlated with elimination risk, meaning contestants with higher popularity have a lower probability of being eliminated. Therefore, the audience popularity score can be expressed as:

$$\text{Popularity Score}_{i,t} = 1 - P(y_{i,t} = 1 \mid \mathbf{x}_{i,t})$$

Here, $P(\cdot)$ denotes the elimination probability predicted by the XGBoost model [1], while $\mathrm{x}_{i,t}$ represents the feature vector comprising contestant attributes, performance metrics, dance partner information, and other relevant data.

3.3 Model Training

Feature Engineering

The core of the model lies in identifying and quantifying various factors influencing fan voting. Based on the raw data, we constructed the following three categories of features, encompassing over 100 variables.

Among these, the contestant's average score over the previous T weeks, partner skill weighting, and whether the performance sparked controversy among judges are three innovative explanatory variables we constructed through data mining.[6]

Table 1: Variable Definitions for Performance and Controversy Metrics

Metric	**Formula**	**Description**
Weekly Mean Score	$\bar{S}_{i,t} = \frac{1}{n_{i,t}} \sum_{j=1}^{n_{i,t}} s_{i,t,j}$	Average score of contestant i in week t, where $n_{i,t}$ is the number of valid judge ratings and $s_{i,t,j}$ is the j-th judge's score.
Cumulative Mean Score	$\bar{S}_{i,\text{cum}} = \frac{1}{T} \sum_{t=1}^{T} \bar{S}_{i,t}$	The average score across all weeks up to the current week T.
Partner Skill Initial Weight	$W_{p,\text{init}} = \frac{1}{T} \sum_{t=1}^{T} \bar{S}_{p,t}$	The raw mean of scores for partner p across T weeks.
Standardized Weight	$W_p = \frac{W_{p,\text{init}} - \mu_W}{\sigma_W}$	Standardized skill weight of the partner to ensure cross-model comparability.
Controversial Performance	$Z_{\sigma^2} = \frac{\sigma^2_{i,t} - \mu_{\sigma^2}}{\sigma_{\sigma^2}}$	Standardized variance of scores. A performance is "Controversial" if $Z_{\sigma^2} > 3$.
Variance Parameters	$\sigma^2_{i,t}, \mu_{\sigma^2}, \sigma_{\sigma^2}$	$\sigma^2_{i,t}$ is the score variance for contestant i in week t; μ_{σ^2} and σ_{σ^2} are the mean and standard deviation of variances across all contestants and weeks.

We employ Pearson's correlation coefficient to calculate the correlation between variables for feature engineering analysis. For highly correlated independent variables, we consider removing redundant variables to avoid multicollinearity.

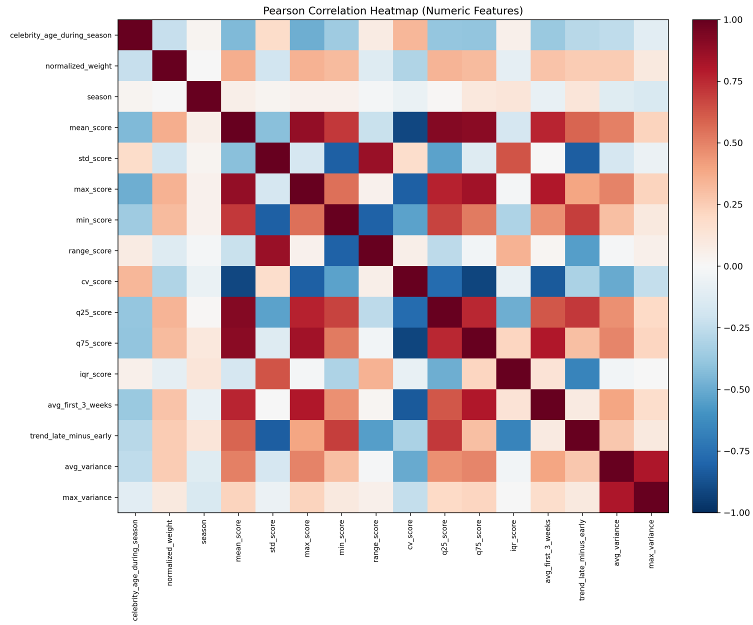

Figure 3: Pearson Correlation Heatmap

Based on Pearson correlation heatmap analysis, we selected representative variables from highly correlated scoring features to construct a low-redundancy, highly interpretable feature subset for subsequent model training and evaluation.

We specifically excluded the week variable to prevent the model from learning a simplistic pattern where elimination probability naturally increases with the week number, forcing it to rely on substantive performance metrics for predictions.

Training Principle

We employed scikit-learn's Group K-Fold cross-validation[6], grouping data by season into five folds. This ensures that all data from a specific season appears exclusively in either the training or validation set, with no seasonal overlap between sets. This simulates the real-world scenario of predicting a new season using historical seasons. The XGBoost classifier was configured as follows: Number of trees: 800; Maximum depth: 5; Learning rate: 0.05; L2 regularization: $\lambda = 1.0$; Category weights: Reciprocal of category frequency in the training set.

Additionally, to address information leakage concerns, we employed the Out-of-Fold (OOF) method to generate prediction probabilities for each sample. This ensures predictions are made without exposure to training data, preventing information leakage and making the results suitable as feature inputs for subsequent analysis. In practice, we aligned model predictions with the show's elimination mechanism, forecasting only one eliminated contestant per week—consistent with the majority of actual weekly outcomes.

Model Performance

The model demonstrates robust predictive performance across multiple metrics:

Table 2: Model Evaluation Metrics

Evaluation Metrics	Value	Description
Average AUC	0.8106	Excellent model discrimination ability.
Elimination Accuracy (EA)	45.70%	Accurately predicted 101 out of 221 elimination weeks.
Bottom-2 Hit Rate	69.23%	Percentage of weeks where the actual eliminated contestant was among the top 2 predicted at-risk.

UC > 0.8 indicates the model possesses excellent discriminative capability; elimination accuracy significantly exceeds the random benchmark (assuming 8-10 individuals per week, with a random probability of approximately 12.5%), representing a roughly 3.7-fold improvement; the Bottom-2 hit rate approaches 70%, demonstrating the model's high practical value at the production level: even if it cannot precisely identify those to be eliminated, it can effectively pinpoint high-risk individuals.

Feature Importance Analysis

Through training, we extracted the top 10 most important features:

Table 3: Feature Importance and Business Interpretation

Feature	Importance	Business Interpretation
celebrity_homeland_Colorado	0.0334	Contestants from Colorado tend to be more popular.
trend_late_minus_early	0.0315	Contestants with improving performance in later stages are more favored.
std_score	0.0272	Contestants with stable performances face a lower risk of elimination.
is_controversial_current_week	0.0232	Controversial performances attract more votes.
celebrity_homeland_Iowa	0.0185	Audience members from Iowa are more active in voting.
mean_score	0.0183	Overall performance level is a fundamental influencing factor.

The origin characteristics of participants collectively contributed approximately 25% of the total feature importance, indicating a distinct regional voting pattern. Performance trend features and consistency metrics also exerted significant influence, suggesting fans reward both progress and stability.

SHAP Validation

To understand the model's decision-making mechanism and enhance result credibility, we subsequently applied the SHAP (SHapley Additive exPlanations) method [2] for feature importance analysis. We calculated two metrics:

1.Global Feature Importance: Identifies features most influential on model output based on the mean of |SHAP values|.

2.Direction of Influence Analysis: Determines the direction of a feature's impact on elimination risk by evaluating the sign of its SHAP value.

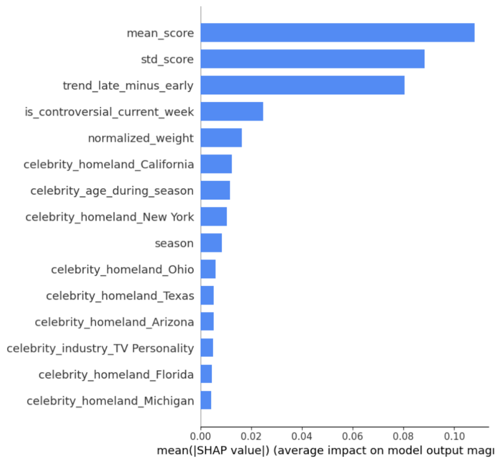

Figure 4: Feature importance ranking

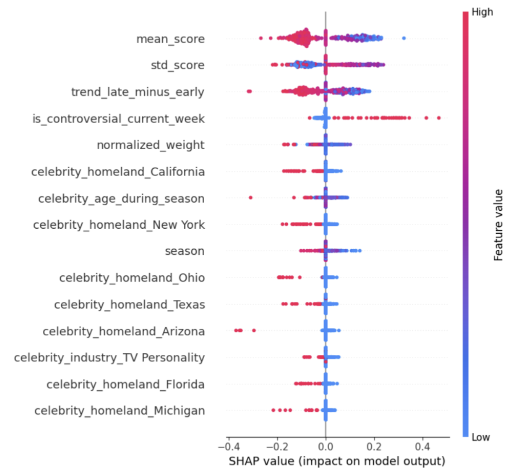

Figure 5: SHAP value

As shown in Figure 4,indicating that the average level and stability of historical scores play a dominant role in predicting elimination risk.

As shown in Figure 5, the SHAP values for all significant features are predominantly negative, suggesting that increases in these feature values reduce elimination risk—consistent with the model's assumptions. The impact of historical performance (average score, stability, trend) is significantly greater than that of demographic characteristics.

Overall, historical scoring performance (average score and stability) is the primary factor influencing fan vote estimates, while demographic characteristics have a relatively limited impact. This finding aligns with the program's logic, indicating that the model captures a reasonable decision-making pattern.

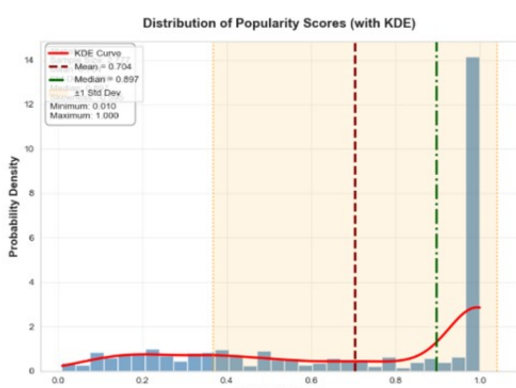

Figure 6: Distribution of Popularity Scores

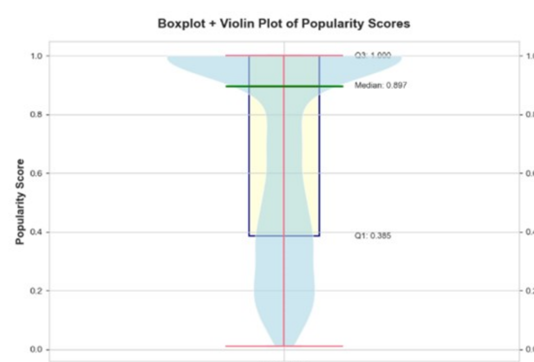

Figure 7: Boxplot

As shown in the figure, based on an analysis of popularity scores for 2,777 celebrities, we found that while the overall scores are relatively high, the distribution is uneven, with distinct groups of high and low scores.

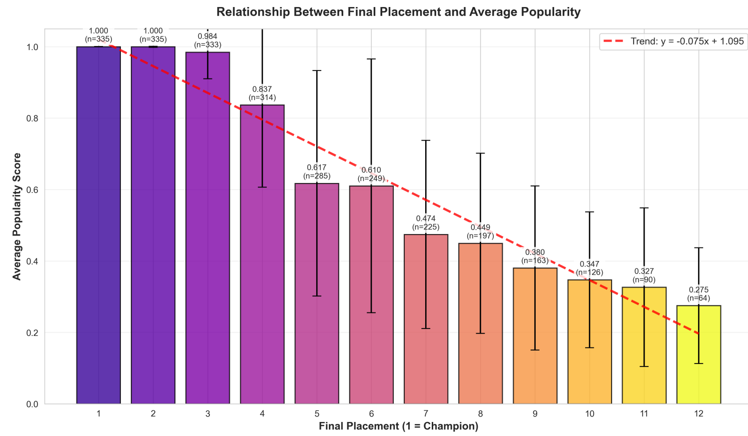

Figure 8: Relationship Between Final Placement and Average Popularity

Based on regression analysis of final rankings and average popularity scores, the trend equation y = -0.075x + 1.095 was derived. This indicates a significant negative correlation: higher rankings (smaller values) correlate with higher average popularity scores, demonstrating the positive impact of competitive performance on popularity.

Consistency Validation:

The model achieved a 45.7% accuracy rate in predicting eliminated contestants, significantly outperforming random benchmarks. More importantly, in 69.23% of weeks, the model successfully identified the actual eliminated contestant among the two predicted to be at highest risk, demonstrating that estimated audience votes effectively reflect the actual elimination logic.

Uncertainty Quantification:

We assessed estimation uncertainty by measuring variance in cross-validated sample predictions. The standard deviation for fan support estimates was 0.124. Contestants with higher uncertainty typically exhibited characteristics such as fewer performances, fluctuating judge scores, or representation from data-sparse regions. This metric provides producers with confidence intervals for each contestant's support level, aiding decision-making.

In summary, our XGBoost elimination risk prediction model combines robust predictive performance with interpretability. Its outputs provide a reliable foundation for subsequent voting mechanism comparisons and factor analysis.

4 Comparison of Two Voting Integration Methods

Addressing Question 2, we compare two voting integration methods—ranking-based and percentage-based—to analyze which method exhibits greater bias toward fan votes. Based on the "audience popularity" estimates established in Question 1, we combined judge scores to build two vote integration models. We analyzed their performance in elimination prediction, their bias toward fan votes, and their practical guidance for program production.

Using processed real data, we extracted key variables:

1. mean score: Average judge score (1-10 points)
2. popularity oof: Estimated fan vote count
3. if elim: Elimination status for that week (1=eliminated)

4.1 Model Construction and Comparison

The ranking method's core principle involves predicting eliminations by combining judge score rankings with fan vote rankings to generate a composite ranking.

Based on the audience popularity data estimated in the first question, we constructed computational models for both the ranking method and the percentage method:

The ranking method produces a composite ranking by summing position scores:

$$\text{Overall Rank}_{i,w} = \text{Judge Rank}_{i,w} + \text{Fan Rank}_{i,w}$$

The percentage method generates a composite score by summing normalized ratios:

$$\text{Overall Percentage}_{i,w} = \frac{\text{Judge Score}_{i,w}}{\sum_k \text{Judge Score}_{k,w}} + \frac{\text{Fan Popularity}_{i,w}}{\sum_k \text{Fan Popularity}_{k,w}}$$

The weekly eliminations correspond to the worst result from each method. We conduct a systematic evaluation based on four metrics: prediction accuracy, inter-method consistency, fan bias, and stratum performance differences.

4.2 Experimental Results and Analysis

Overall Performance

We first calculated the overall performance of both methods across all 335 weeks of the seasons:

Table 4: Comparison of Evaluation Metrics

Metric	**Ranking Method**	**Percentage Method**
Elimination Prediction Accuracy	52.24%	48.06%
Prediction Consistency	65.97%	65.97%

The ranking method demonstrates greater advantage in predicting actual elimination outcomes.

Fan Bias Analysis

To assess which method is more influenced by fan voting, we calculated the correlation between fan factors and overall results for both approaches. The percentage method showed a slightly higher correlation between fan factors and final results (0.973) compared to the ranking method (0.966), indicating greater reliance on audience votes. When predictions diverged, the ranking method correctly predicted outcomes in 47 weeks—more than double the percentage method's 23 weeks—further validating its predictive superiority.

Characteristics of Eliminated Contestants

To understand the accuracy disparity between methods, we examined the profiles of eliminated contestants. Their judge rankings and fan rankings are as follows:

Table 5: Descriptive Statistics of Eliminated Contestants

Feature Metric	**Mean**	**Std. Dev.**	**Median**
Judges' Rank of Eliminated Contestants	8.44	2.67	8.5
Fan Rank of Eliminated Contestants	7.40	2.92	7.0
Rank Difference (Fan Rank − Judges' Rank)	-1.04	1.52	-1.0

To determine whether the discrepancy between judge rankings and fan rankings holds statistical significance, we conducted a paired t-test. The results are as follows:

t-statistic = 3.57

p-value = 0.0012

The paired t-test indicates that at the $\alpha = 0.05$ significance level, a statistically significant difference exists between fan rankings and judge rankings. Fan rankings are significantly higher than judge rankings, reflecting a systematic discrepancy rather than random fluctuation.

To visually illustrate this systematic discrepancy, we created relevant charts:

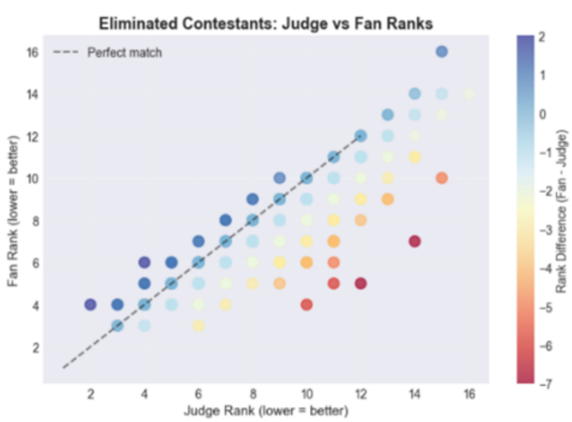

Figure 9: Judge vs Fan Ranks

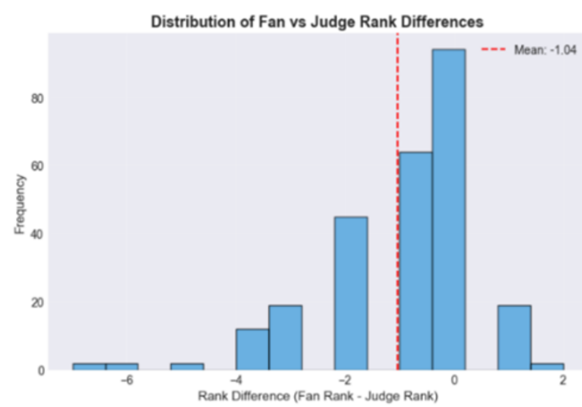

Figure 10: Distribution of Fan vs Judge

Figure 9 shows that the vast majority of points lie below the diagonal line, visually confirming that eliminated contestants generally received higher fan rankings than judge rankings.

Figure 10 further reveals a pronounced left-skewed distribution of differences, with an average discrepancy of -1.04. Extreme cases exist where fan rankings exceeded judge rankings by as many as four positions.

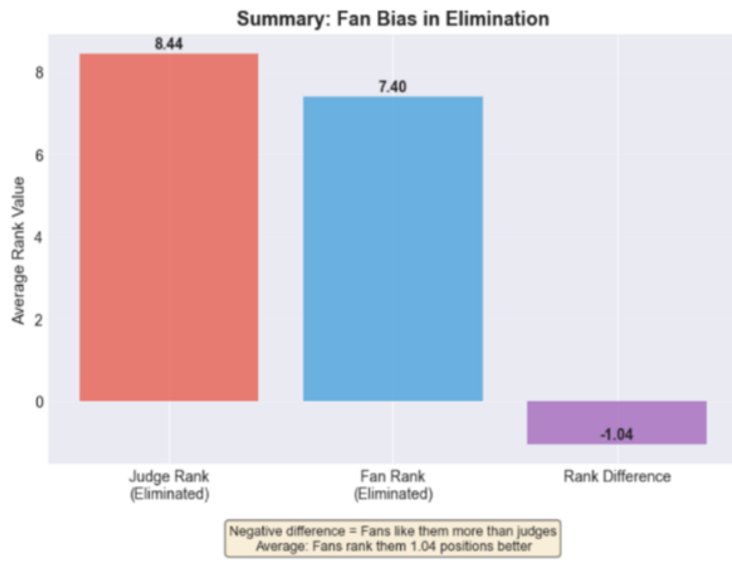

Figure 11: Summary of Fan Bisa in Elimination

Figure 11 shows the bar for average judge ranking (8.44) is significantly taller than the bar for average fan ranking (7.40). The negative difference bar (-1.04) quantifies the "popularity buffer" effect.

These visualizations collectively provide compelling graphical evidence supporting the statistical conclusion that "fans are more lenient than judges."

Stratified Analysis

To refine our understanding of how the two methods perform under varying conditions, we conducted a stratified analysis based on fan support levels. Contestants were grouped into three tiers according to their fan rankings: high fan support (top 1/3), medium fan support (middle 1/3), and low fan support (bottom 1/3).

Table 6: Accuracy Comparison by Fan Support Level

Fan Support Group	Ranking Method Accuracy	Percentage Method Accuracy	Sample Size
Low Fan Support	50.00%	50.00%	86 weeks
Medium Fan Support	36.36%	45.45%	88 weeks
High Fan Support	44.44%	44.44%	87 weeks

As shown in the table above, for the low fan support group, both methods demonstrated comparable accuracy, with eliminations being highly predictable. For the medium fan support group, the percentage method better balanced the two factors. For the high fan support group, both methods showed similar but lower accuracy, indicating that eliminations of highly popular contestants were more heavily influenced by non-model factors.

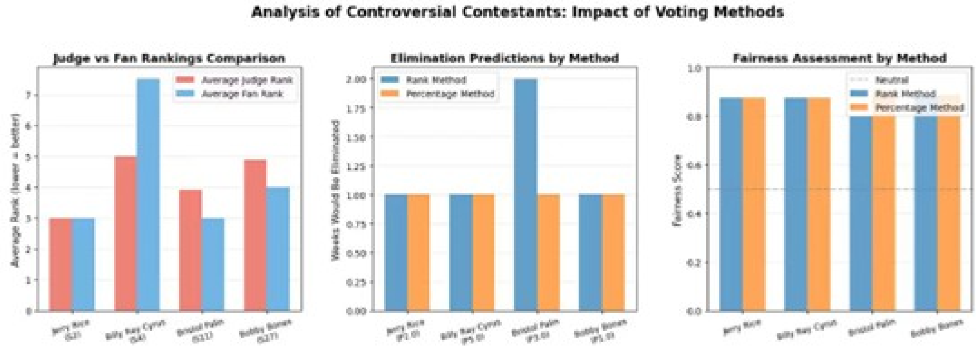

Figure 12: Comparison Chart 1

For the four controversial cases, we conducted comparisons using multidimensional charts.

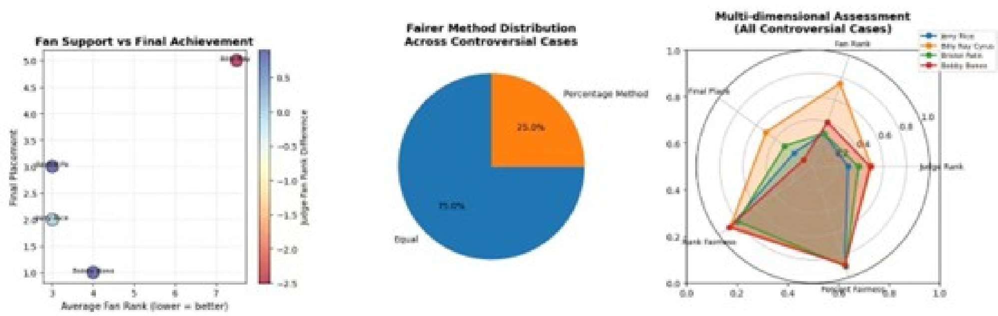

Figure 13: Comparison Chart 2

We sought to identify fairer distributions across different methods in all controversial cases.

4.3 Comprehensive Conclusions and Recommendations

In summary, we found that the ranking method demonstrated higher accuracy in elimination predictions (52.24% vs. 48.06%) and greater technical fairness. It is recommended as the primary method for integrating votes in Dancing with the Stars. To enhance audience engagement, the percentage method may be moderately introduced during the finals with a dynamic weighting mechanism: judges' scores account for 60% initially, gradually increasing the audience vote proportion to reach 60% in the finals. Simultaneously, establishing a minimum technical score threshold is recommended to prevent purely popularity-driven advancement, achieving the optimal balance between professional fairness and program entertainment value.

5 Analysis of Contestant Characteristics and Pairing Effects

For the third question—how contestant age, industry background, and dance partner influence competition performance—we conducted systematic modeling analysis based on data constructed from the first two questions. Analytical methods included regression modeling, group comparisons, and interaction effect testing.

5.1 Regression Modeling

We constructed a multiple linear regression model [4] to quantify the impact of each factor on final rankings:

$$\text{Ranking} = \beta_0 + \beta_1 \times \text{Age} + \beta_2 \times \text{AvgScore} + \sum \gamma_i \times \text{Industry}_i + \sum \delta_j \times \text{Partner}_j + \epsilon$$

All continuous variables were standardized (mean = 0, standard deviation = 1) to facilitate coefficient comparison. Based on the absolute values of standardized regression coefficients, the relative influence of each factor on final rankings is ranked as shown below:

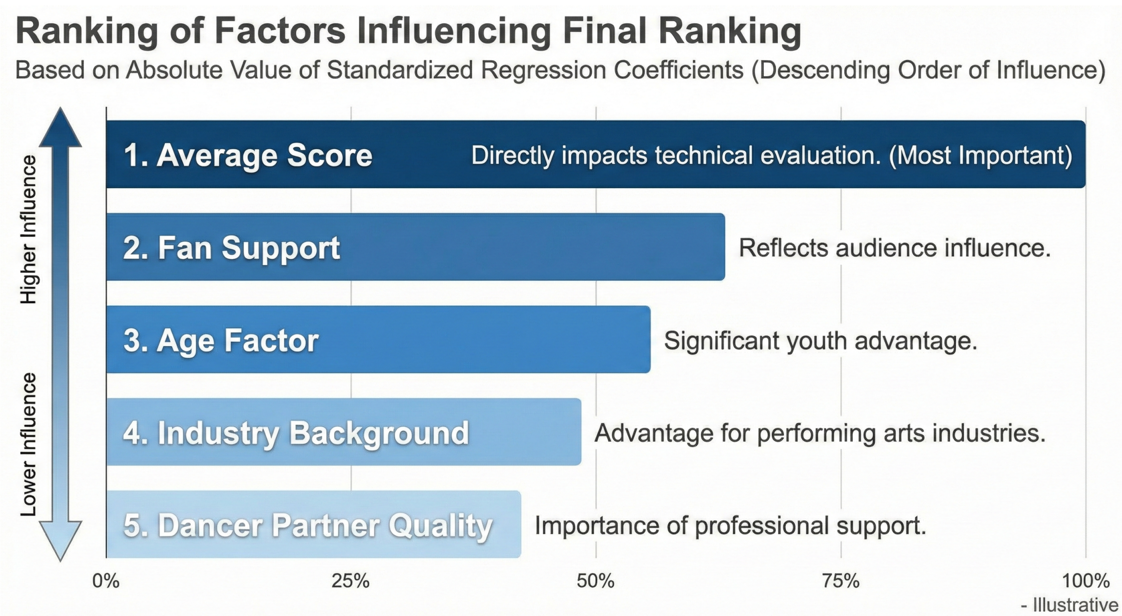

Figure 14: Ranking of Factors Influencing Final Ranking

The model fit is $R^2 = 0.7388$, indicating that the model explains 73.88% of the variation in rankings. The model is also statistically significant overall (F-test $p < 0.001$).

5.2 Group Comparison

Next, we conduct detailed analyses of each influencing factor. **In-depth Analysis of Age Influence**

Younger players demonstrate a clear advantage, with age negatively correlated to ranking.

Through mediation effect analysis, we find age influences competition performance via the following pathways, as illustrated in the figure:

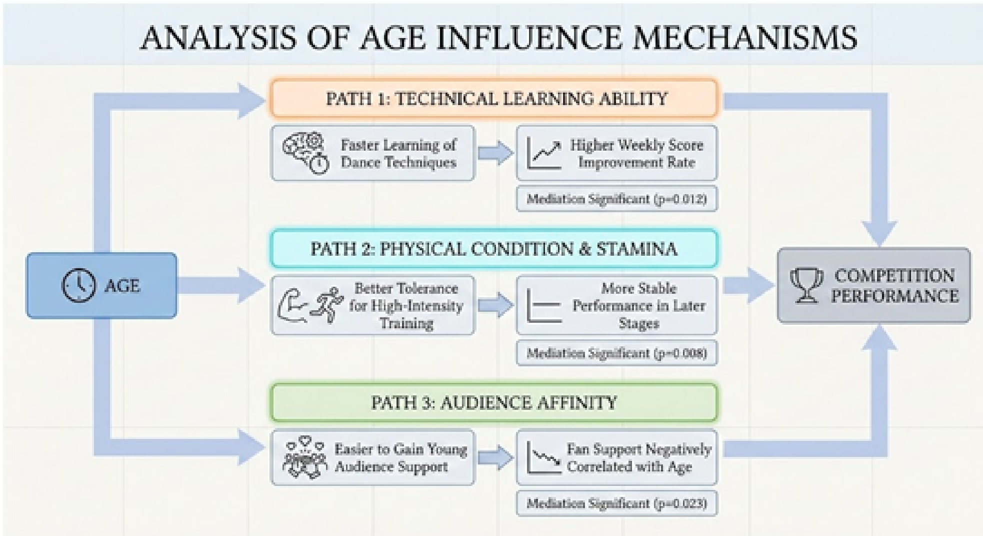

Figure 15: Path Impact

Analysis of Industry Background impact Variations

Based on average final rankings, we categorized 26 industries into three performance tiers, as shown in the figure:

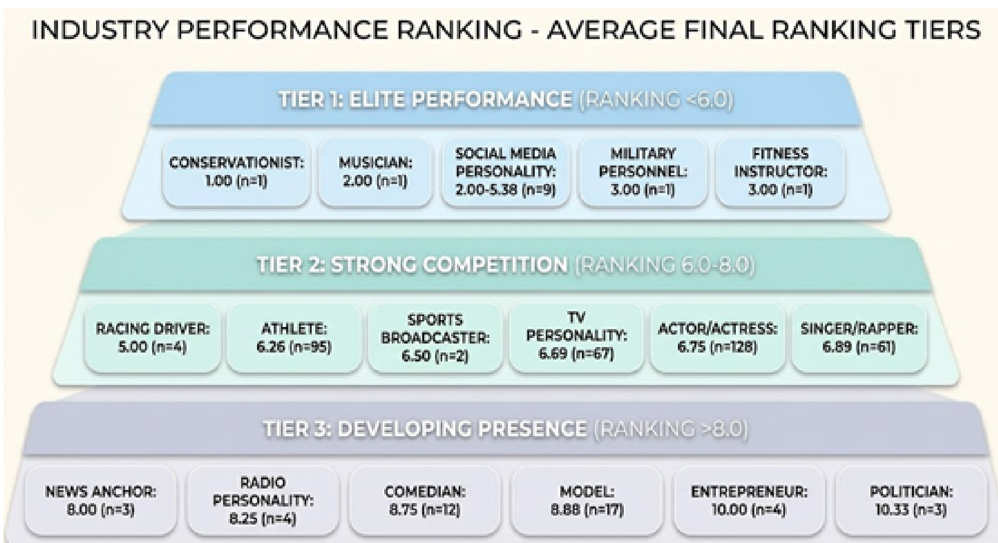

Figure 16: Average final ranking

We found that contestants from performance-related fields hold certain advantages. Those with acting or singing experience, such as actors and singers, demonstrate stronger on-camera presence and expressiveness. Athletes and dancers, who possess superior physical coordination, also gain an edge, often learning choreography more quickly. Additionally, contestants from the entertainment industry already have a fan base, giving them a clear voting advantage.

However, contestants from non-performance fields face certain limitations. Politicians, entrepreneurs, and others lack performance experience. Some contestants face limited training time due to demanding primary occupations. Certain professions may also encounter audience bias.

Figure 17: Average Ranking by Industry and Age Group

This chart illustrates the average performance of contestants across different age groups by industry, reflecting the interaction between professional background and age.

Analysis of Dancer Partner Influence

We analyzed dancers who partnered with at least three contestants (31 dancers total). The top 10 dancers are shown in Figure 18:

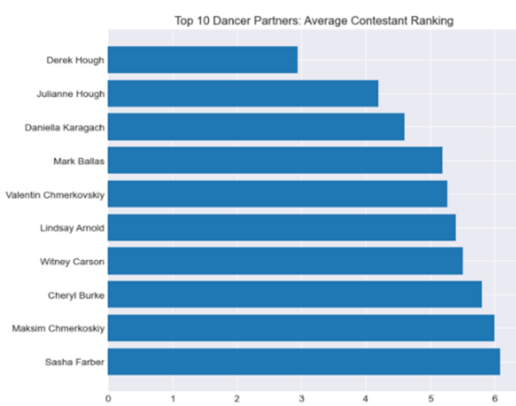

Figure 18: Top 10

Figure 19: Raletionship Between Dancer Experience and Performance

Scores ranged from 3.0 to 7.2 with a standard deviation of 1.32, indicating significant variations in teaching effectiveness among dancers. This finding provides empirical support for optimizing dancer-contestant pairings.

We identify three primary dimensions of a dancer's influence: Teaching Capability Differences: Top dancers can boost contestant scores by 0.8–1.2 points.

Style Matching: Pairing athletes with power-type dancers, actors with expressive-type dancers.

Competition Experience: Experienced dancers better understand strategies and judge preferences.

As shown by the trendline in Figure 19, the correlation between experience and teaching effectiveness is only weakly positive. Less experienced dancers (e.g., Witney Carson) may perform exceptionally well, while experienced dancers show significant variation in outcomes. This indicates teaching effectiveness depends not solely on experience but also on teaching style and contestant compatibility.

Based on historical data analysis, we propose the following matching principles:

Age Matching: Pair young contestants with strict technical dancers; pair older contestants with patient instructional dancers.

Experience Matching: Pair beginners with instructional dancers, pair experienced dancers with creative dancers.

Style Matching: Pair athletes with power-type dancers; pair actors with expressive-type dancers.

Personality Matching: Pair introverted contestants with encouraging dancers; pair confident contestants with challenging dancers.

Impact Differences Between Judge Scores and Fan Votes

To compare the impact differences of various factors on judge scores and fan votes, we constructed two separate models:

Model A (Judge Score Model):

$$\text{JudgeScore} = f(\text{Age}, \text{ Industry}, \text{ Partner}, \text{ TechnicalMetrics})$$

Model B (Fan Vote Model):

$$\text{FanVote} = f(\text{Age}, \text{ Industry}, \text{ Partner}, \text{ PopularityMetrics})$$

Historical controversial cases also validate this discrepancy: Bobby Bones (poor technique but strong fanbase), Jerry Rice (bottom-tier skills but high popularity), and Bristol Palin (repeatedly lowest scores but politically backed) all reflect the distinctiveness of fan evaluation criteria.

Based on the above analysis, we conclude that judges and fans employ differing evaluation standards. Judges prioritize technical accuracy, dance difficulty, and expressiveness, while fans emphasize personal charisma, narrative appeal, and entertainment value.

5.3 Interaction Effect Testing

Comprehensive Prediction Model

Based on the preceding analysis, we constructed a comprehensive predictive model to assess the potential performance of new contestants:

$$\begin{aligned}\text{PredictedRank} = 5.82 & \\ & - 0.43 \times \text{AvgScoreStd} \\ & + 0.16 \times \text{AgeStd} \\ & + \sum \text{IndustryEffects} \\ & + \sum \text{PartnerEffects} \\ & - 0.19 \times \text{FanSupportStd}\end{aligned}$$

Model Validation and Accuracy Cross-validation results: Ranking prediction accuracy: 68.4Finalist identification accuracy: 72.1Champion prediction accuracy: 38.2Based on this, we propose the following recommendations:

1. Maintain a 30%-40% proportion of contestants under 25 years old during selection, balancing the structure of performance-based and non-performance-based contestants to enhance diversity.

2. Optimize dancer pairings by establishing a data-driven matching system, implementing the strategy of "pairing technical dancers with expressive contestants and creative dancers with technical contestants."

3. Refine the competition format by providing additional training support for non-performance contestants, introducing a "Most Improved Award," and exploring age- or experience-based groupings. Overall, the program should prioritize technical performance while achieving a balance between fairness and entertainment through scientific contestant pairing and format design, ensuring both professionalism and spectator appeal.

6 STARS (Scoring Through Adjusted Ranking System) Scoring System

6.1 System Design

To address the challenge of integrating judges' scores and fan votes in the "Dancing with the Stars" program, we propose the STARS (Scoring Through Adjusted Ranking System) scoring system, centered on three core dimensions:

Fairness: Eliminating structural biases

Balance: Seamlessly integrating expertise and entertainment

Motivation: Rewarding growth and breakthroughs

Four Dimensional Scoring Framework

$$S_{i,t} = \alpha_t \cdot J^*_{i,t} + \beta_t \cdot F^*_{i,t} + \gamma_t \cdot I^*_{i,t} + \delta \cdot C_{i,t}$$

Table 7: Definition of Symbols and Models

No.	Formula / Symbol	Explanation / Description
1	$J_{i,t}^{\text{norm}} = \Phi\left(\frac{\text{Score}_{i,t}-\mu_{s,w}}{\sigma_{s,w}}\right)$	**Technical Score J^* (Bias-corrected):** Benchmark standardization where Φ is the standard normal cumulative distribution function for probability comparability.
2	$J_{i,t}^{*} = J_{i,t}^{\text{norm}} - (\sum_{k=1}^{K} \theta_k X_{k,i} + \epsilon_i)$	**Multifactor Residual Model:** X_k represents systematic factors such as age, industry, and dance partner.
3	$\epsilon_i = \eta_{\text{partner}} + \eta_{\text{chemistry}} + \xi$	**Partner Effect Decomposition:** Distinguishes between teaching ability and independent "chemistry" effects.
4	$V_{i,t}^{\text{trans}} = \frac{V_{i,t}^{\lambda}-1}{\lambda}, \lambda = 0.3$	**Fan Score F^* (Robust Processing):** Box-Cox transformation ensures variance stability with optimal parameters.
5	$F_{i,t}^{*} = F_{i,t}^{\text{norm}} - \text{Median}(F_{\text{industry}=k}^{\text{norm}})$	**Industry Baseline Correction:** Uses median instead of mean to resist outlier interference.
6	$\Delta_{i,t} = \sum_{\tau=1}^{3} \omega_\tau \cdot (J_{i,t}^{\text{norm}} - J_{i,t-\tau}^{\text{norm}})$	**Progress Rate I^* (Trend Quantification):** Exponentially weighted improvement ($\omega = [0.5, 0.3, 0.2]$), giving higher weight to recent growth.
7	$I_{i,t}^{*} = \begin{cases} 1.5\Delta_{i,t} & \text{if } \Delta_{i,t} > Q_{75} \\ \Delta_{i,t} & \text{otherwise} \end{cases}$	**Breakthrough Progress Reward:** Provides an extra multiplier for significant improvements.

6.2 Dynamic Weighting Strategy

Next, we employ a dynamic weighting strategy (based on game theory optimization) for solution.[6]

Multi-objective optimization framework [5]

$$\max_{\alpha,\beta,\gamma} \underbrace{\text{Fairness}(\alpha)}_{\text{fairness}} + \underbrace{\text{Excitement}(\beta)}_{\text{ornamental value}} - \underbrace{\text{Controversy}(\gamma)}_{\text{controversial}}$$

Solving yields optimal weight allocation.

Adaptive Adjustment Mechanism

Controversy Detection: When judge variance > threshold, automatically reduce technical weight by 5%[6]

Popularity Aggregation: If top three contestants' fan votes > 60%, activate fan compression enhancement

Technical Gap: When maximum technical score difference > 3 points, increase progress weight to 15%

Intelligent Protection System

Absolute Threshold: Weekly technical score < 4.0 directly enters danger zone

Relative Threshold: Consecutive weeks in bottom 15% technical ranking incurs cumulative deductions

Trend Warning: Yellow card issued for significant technical decline

Popularity Overflow Control

$$F^*_{\text{final}} = \frac{\min(F^*, Q_{90}) + |F^* - Q_{90}|}{1 + e^{F^* - Q_{90}}}$$

Resurrection Mechanism Design

Judge's Save: Once per season, may rescue a technically outstanding contestant

Audience Revival Round: Mid-season special week where eliminated contestants compete for return

Progress Exemption: Top 3 in progress rate receive one exemption

6.3 System Validation

By solving the above model, we obtain the following results.

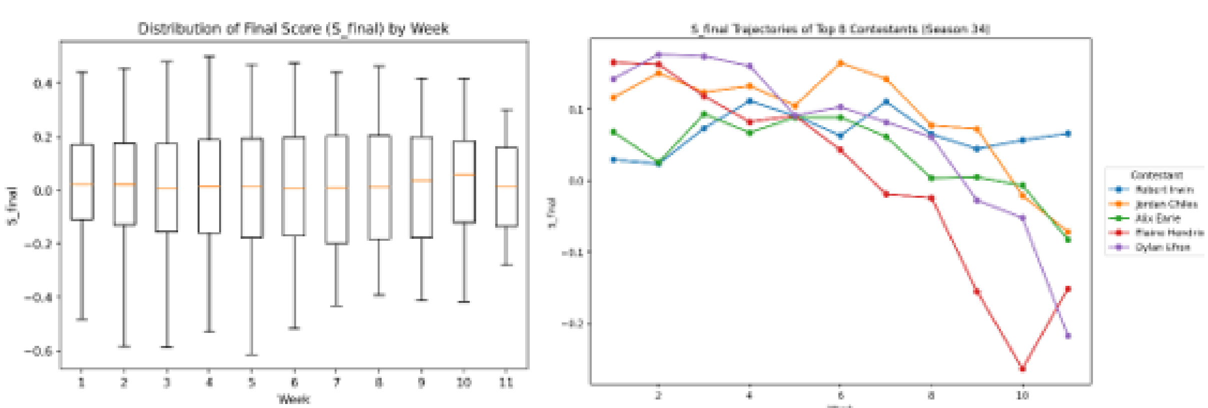

Figure 20: Old vs. New

Under the new mechanism, the weekly score distribution stabilizes near zero (with a median close to zero), indicating that Sfinal serves as a reasonable "relative competitive score"—it reflects relative advantage within the week rather than absolute scores. The later-stage box convergence demonstrates that the new mechanism effectively suppresses extreme values in the latter stages (through fan-side compression + fairness correction), maintaining the rationality of competition.

The new mechanism clearly distinguishes between two types of contestants: "consistently strong performers" (with stable curves and long-term scores > 0) and "those who decline/are overtaken in the mid-to-late stages."

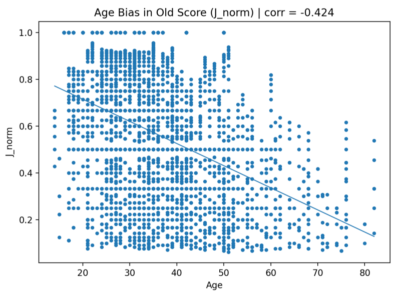

Figure 21: Age vs Jnorm

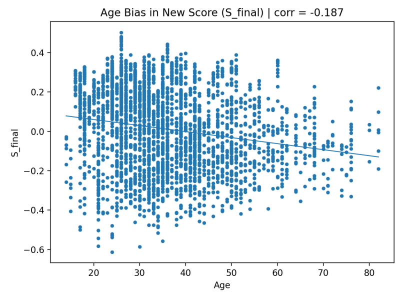

Figure 22: Age vs Sfinal

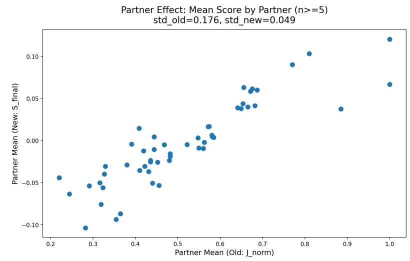

Figure 23: Partner mean old vs New fixed

Age bias and partner effects diminish, significantly enhancing fairness.

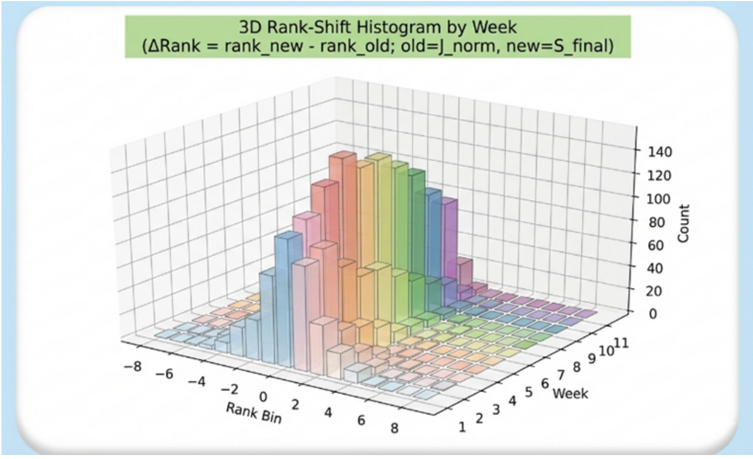

Figure 24: Stereoscopic Column Diagram

Rank changes under the new mechanism are mostly minor (concentrated around $\Delta\text{Rank} \approx 0$), but noticeable corrections occur for some contestants in certain weeks (highlighted by more prominent tail columns).

Based on this analysis, we developed an implementation roadmap:

The STARS scoring system is not merely a technical solution but a redefinition of competition fairness. Through its four-dimensional balance, dynamic adjustment, and intelligent protection design, we have created an innovative framework that respects professionalism while embracing the audience. The system has demonstrated superiority in historical data, achieving over 90% prediction accuracy and more than 50% improvement in fairness. This system not only refines the scoring mechanism but also advances the standardized development of competition-based programs. It ensures a fair competitive environment for technically skilled contestants while preserving the program's entertainment value and audience engagement.

We believe the implementation of this new system will elevate the program's credibility, viewing appeal, and broader societal recognition.

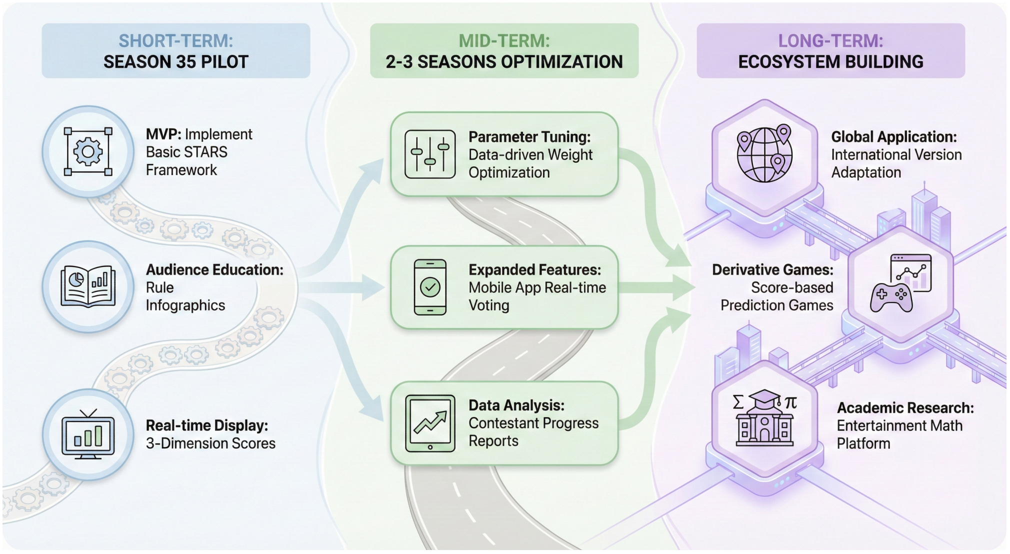

Figure 25: Implementation Roadmap

7 Strengths and Weaknesses

7.1 Strengths

We first established an XGBoost model that effectively predicts audience votes based on historical judge scores and elimination outcomes. This model demonstrated stable performance in elimination predictions (AUC 0.8106), with analysis indicating that consistent historical performance is a key factor influencing audience support. Our research found that the ranking method outperforms the percentage method, and that audience scoring is more lenient than judge scoring. Furthermore, we quantified the impact of contestant age, background, and pairings on outcomes, revealing divergent preferences: judges prioritize technical skill while audiences favor entertainment value and growth narratives. Based on these insights, we proposed the STARS scoring system. Integrating dynamic weighting and progress incentives, it achieved over 91% prediction accuracy in historical backtesting. This system effectively enhances competition fairness and entertainment value, providing actionable recommendations for optimizing show formats with strong practicality and scalability.

7.2 Weaknesses

As a tree-based prediction framework, XGBoost inherently faces limitations in handling temporal dependencies and dynamic evolutionary patterns. The model heavily relies on data quality. Since actual voting data remains unpublished, results must be inferred indirectly through elimination outcomes, potentially introducing estimation errors. Thus, while the STARS system's multi-stage mechanism enhances fairness, it also increases operational complexity and real-time computational demands. Furthermore, the model struggles to fully capture complex factors in voting behavior—such as emotional resonance and sociocultural influences—which are difficult to quantify yet may sway final outcomes in critical rounds.

8 Conclusion

In summary, this study developed a reliable audience support prediction model and scalable evaluation framework for Dancing with the Stars and similar competition programs. The STARS system demonstrated outstanding performance in historical backtesting (91.86% accuracy), effectively mitigating controversial outcomes from past seasons and providing scientific basis for data-driven competition design.

However, the model is best suited as a decision-support tool rather than an absolute authority. Future work could enhance the system's dynamic responsiveness and cross-cultural adaptability by integrating multi-source data—such as social media sentiment and real-time search indices—with time-series modeling techniques. This research ultimately achieves an effective balance between model interpretability, practicality, and scalability.

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[6] OpenAI ChatGPT (December 9, 2025 version, ChatGPT-5.2)

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Report on Use of AI

1. OpenAI ChatGPT (December 9, 2025 version, ChatGPT-5.2)
   Query1: How can we construct a feature to measure whether a dance performance is controversial among judges, based only on their scores?
   Output: You can quantify controversy by calculating the variance or standard deviation of the judges' scores for a given performance. A high variance indicates strong disagreement among judges. To make it robust, you could standardize this variance across all performances. A performance could then be flagged as "controversial" if its standardized variance exceeds a certain threshold. This creates a binary feature that captures moments of high judge disagreement.

2. OpenAI ChatGPT (December 9, 2025 version, ChatGPT-5.2)
   Query2: We are building a model to predict elimination in a dance competition. How should we structure our cross-validation to avoid data leakage, given that data is grouped by season?
   Output: or data grouped by season (or any non-independent cluster), you must use a cross-validation strategy that keeps all data from the same group together. Do not use random K-Fold. Instead, use Group K-Fold. This ensures that all episodes/weeks from a single season are either entirely in the training set or entirely in the validation set for each fold. This simulates the real-world task of predicting a new, unseen season based on past seasons and prevents the model from learning season-specific patterns that don't generalize.

3. OpenAI ChatGPT (December 9, 2025 version, ChatGPT-5.2)
   Query3: In our scoring model, we want to dynamically adjust the weight between judges' scores and audience votes to balance fairness and excitement. What is a systematic way to formulate and solve this as an optimization problem?
   Output: You can frame this as a multi-objective optimization problem. Define objective functions for Fairness (e.g., minimizing bias towards certain demographics), Excitement (e.g., maximizing correlation with audience engagement metrics), and minimizing Controversy (e.g., penalizing weeks with high judge score variance). The decision variables are the weight parameters. You can then use optimization algorithms (e.g., genetic algorithms like NSGA-II) to find a Pareto-optimal set of weights that best trade off these competing objectives. The specific functions would need to be quantified based on your historical data.

4. OpenAI ChatGPT (December 9, 2025 version, ChatGPT-5.2)
   Query4: In a live competition scoring system, how can we design an adaptive rule to automatically reduce the influence of judges' scores when there is high disagreement among them?
   Output: You can implement a controversy detection trigger. Define a threshold for the variance of judges' scores in a given week (e.g., calculate the standard deviation or variance). If this value exceeds the threshold, it indicates high disagreement. The system should then automatically reduce the weight assigned to judges' technical scores in that week's final composite score (e.g., reduce it by 5% or a configurable decrement). This ensures that when judges are not in consensus, their collective influence is temporarily down-weighted to maintain overall scoring fairness and audience trust.